

## Exercise 1: Distribution and Concentration

Consider the following distribution of income in a fictional society of 11 people: {23, 1, 11, 24, 35, 219, 26, 8, 14, 31, 85} (in monetary units).

- a) Determine mean, median and Gini coefficient for the data.

*See Gini coefficient Excel in ISIS*

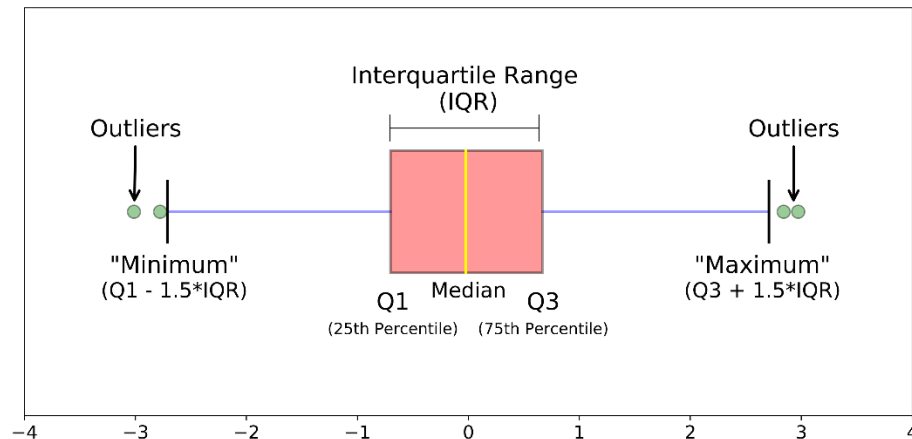
- b) Sketch out a plot that lets you see the concentration of income with ease. To do so, first sort the data. Do not forget to properly label your diagram.

One possibility:



- c) Draw a box plot based on the above data. Include bottom and top outliers, if necessary. Hint: The whiskers extend the box by a length of 1.5 times the interquartile range. Do not forget to label all elements and axes of your plot.

*A boxplot displays the locality, spread, and skewness of our data. An exemplary boxplot can look like this:*



The center point (yellow line) of a boxplot is the median value (in our case 24). The “box” covers our 25th (Q1) to 75th (Q3) percentile of our data. In our case, we can get the Q1 and Q3 points by looking at the median values “left” and “right” of our overall median, in our case:

1, 8, **11**, 14, 23, **24**, 26, 31, **35**, 85, 219

Overall, we get:

- Median: **24**
- Q1: **11** (“left median”)
- Q3: **35** (“right median”)

The Interquartile Range (IQR) is calculated as the distance between Q1 and Q3 – in our case:

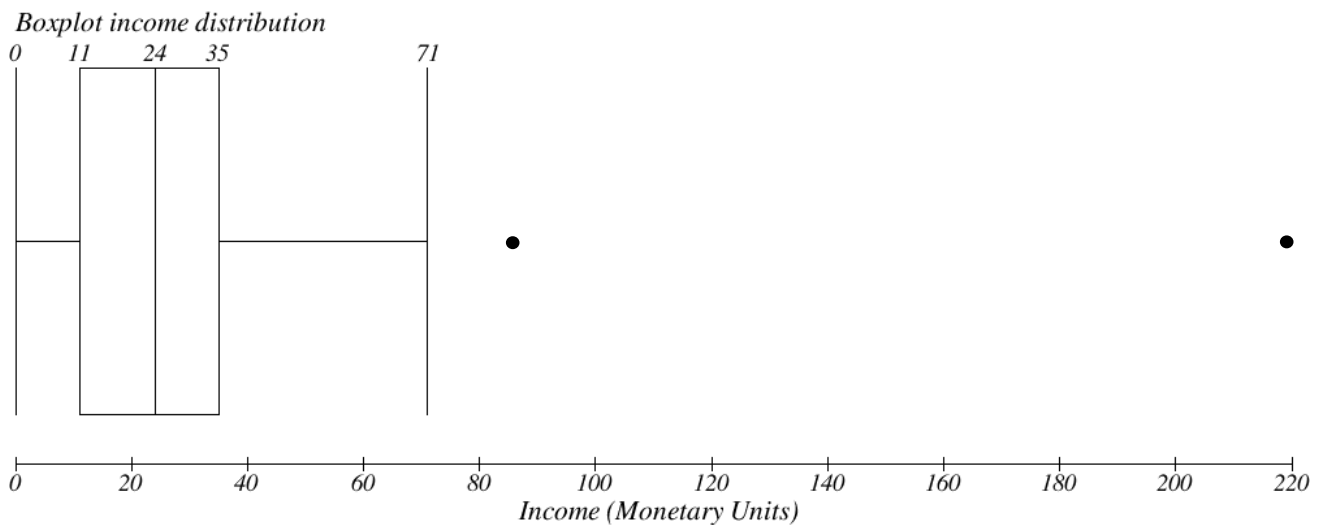
$IQR: 35 - 11 = 24$

The “whiskers” of a boxplot extend beyond the Q1 and Q3 marks and cover “Minimum” and “Maximum” values in our data. However, whiskers can only be  $1.5 \cdot IQR$  long (in our case,  $1.5 \cdot 24 \rightarrow 36$  units). Values beyond the whiskers are called “outliers”. In our case, we are measuring income – as such, no value below 0 is possible and the whiskers of the boxplot end at 0. For our Maximum, the whiskers only cover the following values:

$Q3 + 1.5 \cdot IQR \rightarrow 35 + 1.5 \cdot IQR \rightarrow 35 + 36 = 71$

As such, we have two outliers beyond Q3  $\rightarrow$  85, 219

Now that we have all the relevant data points to plot our boxplot diagram, this is how it looks like:



## Exercise 2: Questionable Visualizations

Consider the following examples of data visualizations and discuss potential shortcomings (you will find the images on the last pages of this document). What would you do differently to improve the visualizations?

*See discussion in tutorial.*

## Exercise 3: Data Visualization

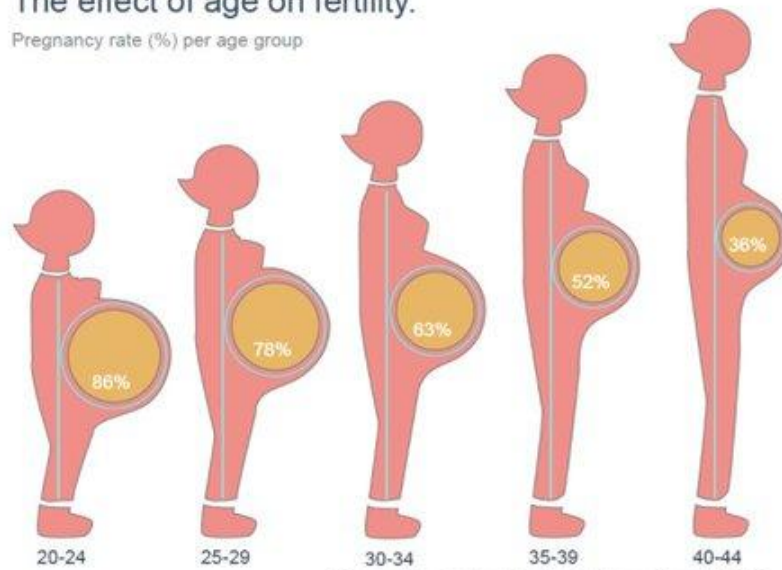
Consider the dataset of TU Berlin researchers `tub_scholars.csv` (as per Google Scholar; early 2022). In this dataset, you will find information on the scholars' presumed gender, their respective number of citations, and up to five thematic tags (e.g. "machine learning", "urban mining", "crowd sourcing"). Visualize this data. Use the TU website to come up with a color scheme for your visualization.

*See RScript in ISIS*

## Exercise 2: Images

### The effect of age on fertility.

Pregnancy rate (%) per age group



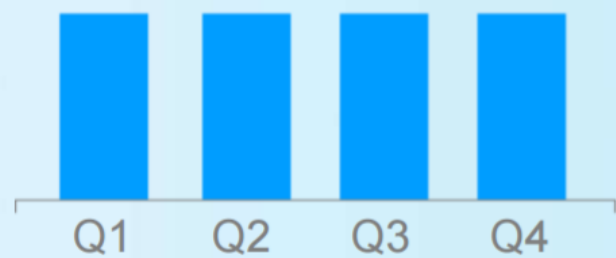
Source: BabyCenter Medical Advisory Board, 2015.

## Increasing Media Continuity

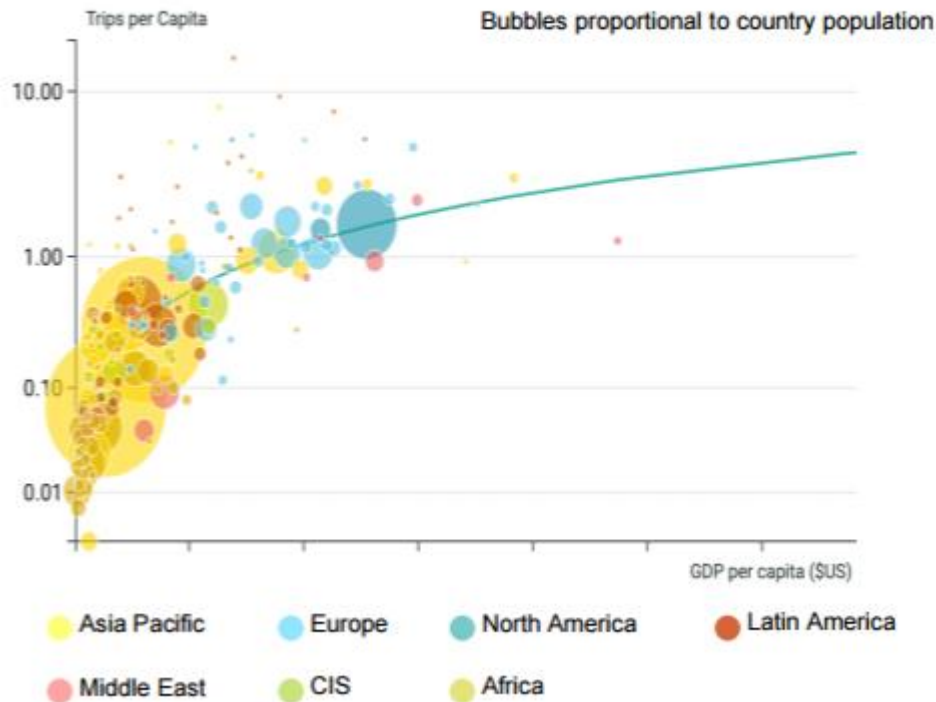
Before



After



## Propensity to travel



### Laws on file

If no colour appears, there is no such law on file

- 2012 election results
- Background check law
- Permit required to purchase
- Licence required to sell
- Records kept on file
- Firearms banned from workplace

### Virginia

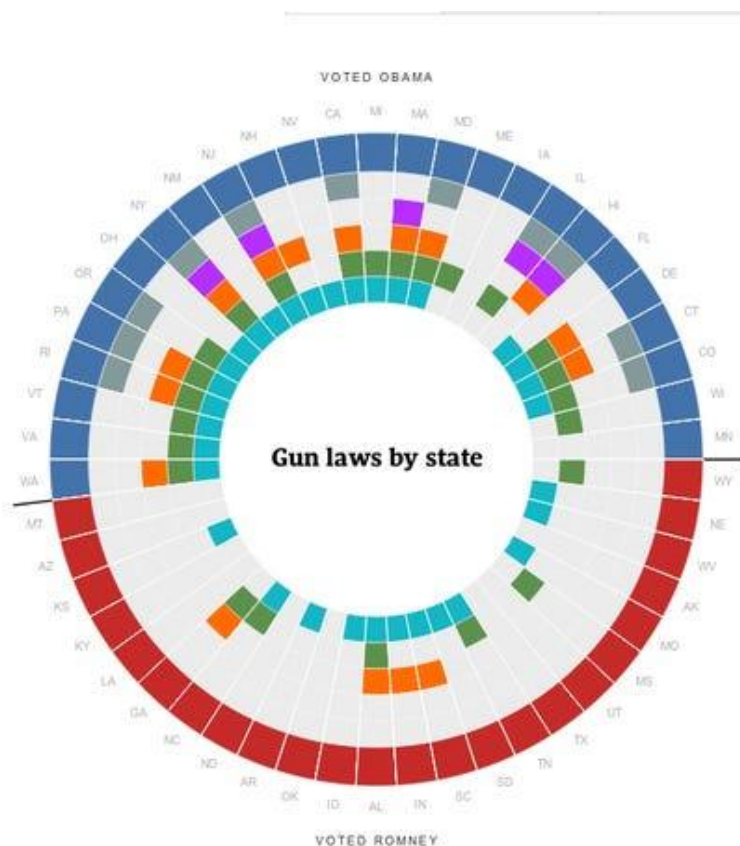
- Voted for Obama in the 2012 election
- Background check:** not required for handguns
- Permit:** not required to buy firearms
- Licence:** not required for dealers
- Records:** kept on file for handgun owners
- Workplace:** firearms not allowed in parking lots

### Overall gun control score: 12

Virginia has a **Brady Campaign score** of 12, which is lower than the national average of 16. The score comes from measuring these and other gun laws according to a weighted points system.

### Murder rate: 2.58

There were 2.58 firearm murders per 100,000 people in Virginia during 2011, which is lower than the national average of 2.77. Overall, it is ranked #27 in murder rates out of 48 states with this data.



## States with the highest firearm murder rate

Louisiana scored only two points on the Brady scale for banning guns from college campuses. It also has the highest firearm murder rate per 100,000 people in the country. Overall, Republican states have an average Brady score of 4.6, compared to 26.73 for states that voted for President Obama in the last election.



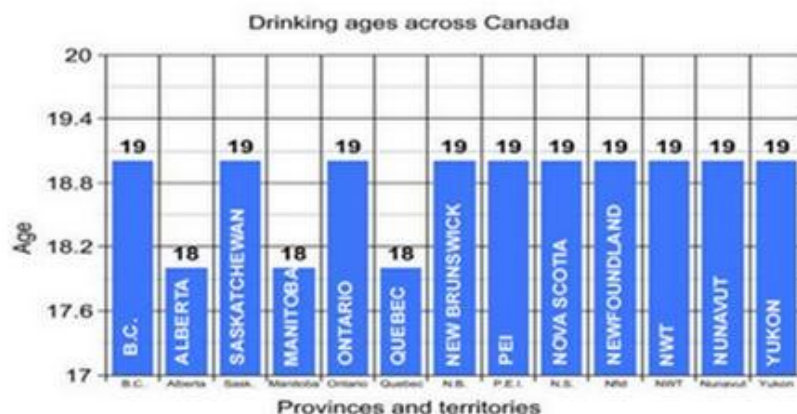
## States with the lowest firearm murder rate

Hawaii has the lowest firearm murder rate in the United States with just 0.07 murders per 100,000 people. South Dakota is the only Republican state to rank on this list. Despite scoring only 7 points on the Brady score and enacting none of the laws highlighted on this chart, Iowa still has one of the lowest firearm murder rates in the country.



# Drinking age will remain 19 in

CBC News Posted: Mar 4, 2013 11:59 AM CST | Last Updated: Mar 4, 2013 11:55 AM CST



Canadian Centre on Substance Abuse

You have to be 19 in Saskatchewan to have a drink, while in Alberta and Manitoba, the drinking age 18. (CBC)

The Saskatchewan Party government has ruled out lowering the drinking age, four months after party members put the issue in the public eye.

